

Why do we care about inverse matrices

• $Ax=b$ b, A are given x is unknown

• checking that $x_1, \dots, x_n$ are linearly independent

solving $\begin{bmatrix} x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix} a = 0$ and $a=0$ is the only sol

• $v \in \text{span}\{x_1, \dots, x_n\}$

$$\begin{bmatrix} x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix} a = v$$

has at least 1 solution

in general linear combination of $\{x_1, \dots, x_n\}$

$$\begin{bmatrix} x_1 & \dots & x_n \\ | & & | \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = a_1 x_1 + \dots + a_n x_n$$

if A has an inverse $x = A^{-1}b$

Every one of our 3 row operations can be represented as a matrix multiplication

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2.4 & 2.5 & 2.6 \\ 7 & 8 & 9 \end{bmatrix}$$

mult row 2 by 7

swap row 1 and 2

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$$

add row 1 to row 3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

row operation $\rightarrow$ matrix mult.

BA resulting in doing a row operation to A:
do the row operation to I and use the result as B

EX: row reduce $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ to I

$$\begin{array}{c} r_1 - r_2 \\ \hline \begin{bmatrix} 1 & 1 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & 0 & 1 & 0 \\ 0 & 0 & 7 & | & 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$\begin{array}{c} \frac{1}{7} r_3 \\ \hline \begin{bmatrix} 1 & 0 & 0 & | & 1 & -1 & 0 \\ 0 & 1 & 0 & | & 0 & 1 & 0 \\ 0 & 0 & 7 & | & 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$\begin{array}{c} A^{-1} \\ \hline \begin{bmatrix} 1 & 0 & 0 & | & 1 & -1 & 0 \\ 0 & 1 & 0 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & \frac{1}{7} \end{bmatrix} \end{array}$$

$$\begin{array}{c} r_1 - r_2 \\ \hline \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \quad \begin{array}{c} \frac{1}{7} r_3 \\ \hline \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{7} \end{bmatrix} \end{array}$$

$$\text{So } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{7} \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{7} \end{bmatrix}}_{A^{-1}}$$

for matrices

$$BA \neq AB \text{ but } A(BC) = (AB)C$$